

Make Your Own IVR with Asterisk

Interactive voice response (IVR) is ubiquitous and now pervades the business and commerce milieu. Using Asterisk, IVR can be easily set up and coded. This fifth article in the series on Asterisk takes a look at how IVR is coded.



Asterisk provides a generic switching platform to run a variety of applications. IVR is commonly used today in most large corporate PBXes. Typically, these are automated voice menus – what you hear when you call a bank or insurance company. The recorded voice will prompt you to input the intended transaction as a choice in the form of digits (DTMF or dual tone multi-frequency tones). The transactions requested for are executed based on the user inputs. In this session, we will look into IVR coding and then the hardware configuration required.

Let's start with a welcome menu, which is a very common feature nowadays in any Asterisk installation. The code is written in the dial plan, which is the central routing control based on pattern matching. The dial plan is generally found in */etc/asterisk/extensions.conf*.

Example 1

- Play the welcome message to the caller
- Ring the extension for 60 seconds
- If unavailable, pass the call to voicemail
- Hang up

Here's the code snippet for this example:

```
[from-pstn]
exten => _,1, Answer();
exten => _,2, Playback(welcome);
exten => _,3, Dial(SIP/${EXTEN},60);
exten => _,4, Voicemail(${EXTEN},u);
exten => _,5, Hangup();
```

[from-pstn] indicates the context in which the call is processed, which is the incoming calls from the PSTN (public switched telephone network – normal PRI or FXO trunk). *exten =>* is a standard keyword to indicate a pattern matching routine. ‘_’ indicates that any extension is matched and the following actions need to be carried out. The second digit ‘1’ after the comma indicates a sequence number. The lines that follow increase the sequence number in ascending order. *Answer()* indicates the call has to be answered so that the voice channels are open in both directions. This is required, so that the users can hear the greetings message and provide their inputs. *Playback(welcome)* instructs the system to search for a file *welcome.gsm* or *welcome.wav* in the default voice directory, and play that file for the user to hear. The file